
Md Saikat Islam Khan

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RESEARCH INTERESTS AND CONTRIBUTIONS

Keywords: Federated Learning, Knowledge Graph Question Answering, Differential Privacy, Explainability.

Research Summary: I am a PhD student in the **BRAINS Lab** at Rensselaer Polytechnic Institute, where my research focuses on advancing **distributed learning** and **explainability**. I develop methods for privacy-preserving and robust training across vertically and horizontally partitioned data, alongside model-agnostic frameworks that generate sparse, actionable counterfactual explanations. Currently, I am extending this work to **complex question answering** over **distributed knowledge graphs**. We aim to support **multi-hop reasoning** and query answering in federated settings without exposing raw entities, relations, or embeddings. More broadly, my research analyzes and reconciles the trade-offs between privacy, robustness, interpretability, and utility in realistic deployment scenarios.

Impact & Recognition: I have published **seven** first-authored papers (**26+** total, including co-authored works) in venues such as CIFER, DAPPS, JKSUCIS, CSBJ, AIMS PH, and ETT. My work has been recognized with one **Best Paper Award** at TCCE. These contributions have garnered significant recognition, with over **2000** citations and an H-index of **15**.

ACADEMIC CREDENTIALS

Rensselaer Polytechnic Institute

Ph.D. in Computer Science

Advised by Dr. Oshani Seneviratne.

CGPA: 3.76 on the scale of 4.00

Troy, New York, USA

August 2023 - May 2028

Rensselaer Polytechnic Institute

M.Sc. in Computer Science

CGPA: 4.00 on the scale of 4.00

Troy, New York, USA

August 2023 - December 2025

Mawlana Bhashani Science and Technology University

B.Sc in Computer Science and Engineering

CGPA: 3.80 on the scale of 4.00

Position: 3rd

Tangail, Bangladesh

February 2015 - December 2019

SELECTED PUBLICATIONS

1. **Khan MS**, Gupta A, Seneviratne O, Patterson S. Fed-RD: Privacy-preserving federated learning for financial crime detection. In 2024 IEEE Symposium on Computational Intelligence for Financial Engineering and Economics (CIFER) 2024 Oct 22 (pp. 1-9). IEEE.
2. Tran L, Chari S, **Khan MS**, Zachariah A, Patterson S, Seneviratne O. A differentially private blockchain-based approach for vertical federated learning. In 2024 IEEE International Conference on Decentralized Applications and Infrastructures (DAPPS) 2024 Jul 15 (pp. 86-92). IEEE.
3. **Khan SI**, Shahrior A, Karim R, Hasan M, Rahman A. MultiNet: A deep neural network approach for detecting breast cancer through multi-scale feature fusion. Journal of King Saud University-Computer and Information Sciences. 2022 Sep 1;34(8):6217-28.
4. **Khan MS**, Rahman A, Debnath T, Karim MR, Nasir MK, Band SS, Mosavi A, Dehzangi I. Accurate brain tumor detection using deep convolutional neural network. Computational and structural biotechnology journal. 2022 Jan 1;20:4733-45.
5. **Khan MS**, Islam N, Uddin J, Islam S, Nasir MK. Water quality prediction and classification based on principal component regression and gradient boosting classifier approach. Journal of King Saud University-Computer and Information Sciences. 2022 Sep 1;34(8):4773-81.

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6. **Khan MS**, Rahman A, Islam S, Nasir MK, Band SS, Mosavi A. IoT and wireless sensor networking-based effluent treatment plant monitoring system. *acta polytechnica hungarica*. 2021 Oct;18(10):205-24.
 7. Rahman A, Hossain MS, Muhammad G, Kundu D, Debnath T, Rahman M, **Khan MS**, Tiwari P, Band SS. Federated learning-based AI approaches in smart healthcare: concepts, taxonomies, challenges and open issues. *Cluster computing*. 2023 Aug;26(4):2271-311.
 8. Nur A, **Khan MS**, Nasir MK. Using fused Contourlet transform and neural features to spot COVID19 infections in CT scan images. *Intelligent Systems with Applications*. 2023 Feb 1;17:200182.
 9. Mazumder B, **Khan MS**, Uddin KM. Biorthogonal wavelet based entropy feature extraction for identification of maize leaf diseases. *Journal of Agriculture and Food Research*. 2023 Dec 1;14:100756.
 10. Rahman MM, **Khan MS**, Babu HM. BreastMultiNet: A multi-scale feature fusion method using deep neural network to detect breast cancer. *Array*. 2022 Dec 1;16:100256.

Under Review

1. **Khan MS**, Seneviratne O, Patterson S. SPICE: Sparse and Proximate Counterfactual Explanations via Feature-Importance-Weighted Perturbation. [Submitted to KDD Conference]
1. **Khan MS**, Seneviratne O. FedKGQA: Privacy-Preserving Federated Learning for Multi-Hop Knowledge Graph Question Answering. [Submitted to KDD Conference]
3. **Khan MS**, Seneviratne O, Patterson S. Fed-RD: Privacy-Preserving Federated Learning for Relational Data. [Submitted to IEEE Transactions on Big Data]

PROFESSIONAL EXPERIENCE

Research Assistant

Aug 2023-Dec 2024

Rensselaer Polytechnic Institute

Funded by CRAFT to investigate efficient, private, and explainable federated learning for financial crime detection.

Lecturer, Computer Science

Jan 2022-Jul 2023

Dhaka International University

Undergraduate Researcher

Jan. 2019 - Dec. 2021

Mawlana Bhashani Science and Technology University

Joined Professor Mostofa's group as an undergraduate student and worked on various projects, such as detecting tumors and breast cancer using DNN models that improved upon past work.

SKILLS

- **Programming Languages:**
 - Comfortable with: Python, Java, JavaScript
 - Have used: C, C++, Haskell
- **Machine Learning Libraries:**
 - Comfortable with: PyTorch, scikit-learn, OpenCV, TorchVision, pandas, NumPy
 - Have used: TensorFlow, Keras
- **Web Frameworks:**
 - Comfortable with: Django, .NET, Flask
 - Have used: React.js, Next.js
- **Environments:**
 - Comfortable with: macOS, Linux
 - Have used: Windows

PROFESSIONAL SERVICE

- Reviewer for major AI conferences, including the **AAAI** Conference and **KDD**.
- Journal reviewer for **IEEE Access** and **Informatics in Medicine Unlocked**.

AWARDS AND ACHIEVEMENTS

- Secured **3rd position** to obtain B.Sc. degree.
- Awarded the **Craft Research Fund Grant** (2023-2024) for conducting research. ([Link](#))
- Won the **best paper award** at *2nd International Conference on Trends in Computational and Cognitive Engineering (TCCE) 2020*. ([Link](#))
- Won the **silver award** at *1st International Poster Competition on Achieving Sustainable Development Goals (SDGs) in Developing Countries*. ([Link](#))
- Took a Workshop on "Leveraging Machine Learning Approach to Software-Defined Network (SDN)" at *IEEE celebration day*. ([Link](#))